# AIBS & AEL Applied Research Handbook

## Chapter 1: Problem Analysis, Target Audience & Research Proposal

*What is actually going wrong for a regional teamwear wholesaler, and who needs to know?*

Team 6 · HAN AIBS/AEL 2026 · v1.1 · SME: 100–200-employee Benelux teamwear wholesaler persona · Sources last appraised 24 Oct 2026

### 1. The question, for this kind of SME

**Main Research Question: How can a regional sports and teamwear wholesaler pragmatically evaluate, select, and deploy AI to improve demand forecasting and stock allocation while managing data-privacy and vendor risk?**

- Which commercial and open-source AI inventory tools fit apparel wholesalers, and how do their data requirements compare?
- Where does AI demand forecasting typically fail on seasonal SME sales cycles?
- How clean and anonymized must ERP data be before an AI agent may touch order processing? Audience: Operations Managers, IT Leads, Warehouse Fulfillment Directors.

### 2. The translation

Regional Benelux teamwear wholesalers (Robey, NOOR, Derbystar, Sportconfex, and the persona wholesaler used throughout this handbook) run centralized 15,000+ m² warehouses holding tens of thousands of SKUs across club kits and training gear, with order volume spiking sharply around KNVB/hockey season registration in August–September (ECI Software Solutions, 2026). Broken size runs, free-text ERP entries (AFAS, Exact, SAP Business One), and a 1–2 person IT team leave no room for custom software. Every research question and AI proposal in this handbook stays anchored to that daily reality, not abstract AI theory.

### 3. Illustrative use case (CRISP-DM)

**Business understanding:** forecast SKU-level demand ahead of the August–September spike. **Data understanding:** profile the wholesaler's own ERP history, including its unstandardized free-text entries. **Data preparation:** anonymize that history before any AI tool touches it. **Modelling:** pilot one core uniform line first, benchmarked against tools from Sub-Question 1. **Evaluation:** judge the pilot against the drift risks from Sub-Question 2, not accuracy alone. **Deployment:** feed validated forecasts into the reorder workflow as a recommendation only — the operations manager keeps sign-off.

### 4. Recommended tooling

| Maturity Tier | Technical Focus | Wholesaler Use Case |
|---|---|---|
| Tier 1: Off-the-shelf | SaaS prompt guardrails | Standardized prompts for club stock queries |
| Tier 2: Workflow automation | Local ERP-scrubbing scripts | Validate size-run completeness before API calls |
| Tier 3: Predictive | RAG + automated reordering | Seasonal reorder alerts tied to registration dates |

### 5. Responsible AI & trust

Because this handbook touches live order data, four rules hold: no AI tool or API sees raw ERP data before anonymization (Sub-Question 3); vendor data-residency and exit terms are checked before any contract; every forecast stays advisory, with a human keeping final sign-off; and any AI-driven stock decision can be explained to the dealers and clubs it affects, never left as a black box.

### 6. First five steps for an owner

- Day 1: map ERP friction points and audit size-run availability across core uniform lines.
- Week 1: restrict external AI/API use to locally anonymized data only.
- Week 2: baseline current cut-to-ship time and catalog update frequency.
- Week 2: set a concrete reduction target for manual order-verification steps.
- Week 3: run short, structured SME interviews to validate assumptions before committing budget.

### 7. sources

Team 6 desk research and sector framing, Week 3 handbook page.
ECI Software Solutions SMB AI survey, ecommercenews.co.nz, 2026, "AI ambition outpaces readiness for smaller businesses."
Australian Government, AI adoption insights, Dec 2025–Feb 2026, ai.gov.au.

# Chapter 2: Geopolitics & Vendor Models

*Which AI vendor landscape should a wholesaler actually trust?*

Team 6 · HAN AIBS/AEL 2026 · v1.1 · SME: 100–200-employee Benelux teamwear wholesaler persona · Sources last appraised 24 Oct 2026

## 1. The question, for this kind of SME

**Main Research Question: Which AI vendor landscape — US hyperscaler, European sovereign, or Chinese industrial platform — can a regional teamwear wholesaler actually trust with order and customer data, without long-term lock-in or breaching EU compliance?**

- How do US, European, and Chinese AI platforms differ on data residency, cost structure, and lock-in risk?
- What source-quality rubric separates audited vendor evidence from marketing claims?
- What technical safeguards must exist before a vendor is allowed to touch ERP or customer data?

## 2. The translation

Unlike a large enterprise, the wholesaler has no dedicated procurement or legal team to vet AI vendors. US hyperscalers (OpenAI, AWS, Anthropic) move fast but bring lock-in, rising API costs, and data-sovereignty exposure; European open-weight models (e.g. Mistral) run on regional infrastructure, meet the EU AI Act and GDPR, and keep margin data local (Reg. (EU) 2024/1689). The standing rule: never adopt a wrapped public API that lacks audited error-rate reporting or that requires uploading unanonymized customer lists.

## 3. Illustrative use case (CRISP-DM)

**Business understanding:** scope the evaluation around the wholesaler's own context — Dutch teamwear, 1–2 person IT. **Data understanding:** run domain-anchored queries — AI demand forecasting for teamwear or apparel wholesalers; size-run ERP integration; EU AI Act compliance for a Dutch manufacturer. **Data preparation:** strip customer PII locally before any source ever reaches an external API. **Modelling:** score every surviving source on a 10-point rubric — relevance 3, methodological transparency 3, institutional authority 2, timeliness/compliance 2. **Evaluation:** exclude any source scoring below 7/10 from the drafting context. **Deployment:** place surviving, audited evidence into the maturity tiers below for a documented go or no-go.

## 4. Recommended tooling

| Maturity Tier | Vendor Architecture | Wholesaler Risk Management |
|---|---|---|
| Tier 1: Off-the-shelf | Proprietary commercial APIs | Enforce acceptable-use policies blocking unanonymized uploads |
| Tier 2: Workflow automation | Hybrid open-weight models (Mistral) | Run local anonymization before external processing |
| Tier 3: Predictive | Self-hosted on-premise RAG | Full data residency, zero vendor lock-in |

## 5. Responsible AI & trust

Vendor trust is a responsible-AI question, not just a procurement one: no unaudited public APIs; EU AI Act and GDPR compliance by default, not by claim; vendor evidence accepted only at 7/10 or above on the rubric above, never on sales copy alone; and data/model export confirmed before signing, so switching vendors later never means starting over.

## 6. First five steps for an owner

- Day 1: run domain-anchored research queries combining teamwear, size-run ERP rules, and EU AI Act compliance.
- Week 1: score every candidate source on the 10-point rubric; disqualify sales pages lacking error rates.
- Week 2: configure local scripts to strip PII and ERP margins before any data reaches an external LLM.
- Week 3: require proof of EU data residency, exportable models, and manual fallback from every shortlisted vendor.
- Week 4: place top candidates on the tier table above and make a documented go/no-go before signing.

## 7. sources

Team 6 desk research on the vendor landscape and SME IT constraints (see Chapter 1).
Regulation (EU) 2024/1689 (EU Artificial Intelligence Act), eur-lex.europa.eu/eli/reg/2024/1689/oj/eng.
Regulation (EU) 2016/679 (GDPR), eur-lex.europa.eu/eli/reg/2016/679/oj.

# Chapter 3: Trust & AI Safety in Seasonal Teamwear Wholesaling

*What can go wrong once AI touches real orders, and how do we stop it?*

Team 6 · HAN AIBS/AEL 2026 · v1.1 · SME: 100–200-employee Benelux teamwear wholesaler persona · Sources last appraised 24 Oct 2026

### 1. The question, for this kind of SME

**Main Research Question: What can go wrong once AI starts touching real orders and client data at a regional teamwear wholesaler, and how can safeguards go in before damage is done rather than after?**

- How does model drift in reorder calculations create stockouts or overstock during seasonal spikes?
- What shadow-AI behavior among staff creates GDPR exposure and intellectual-property leakage?
- What vendor lock-in and compliance exposure follows from choosing the wrong AI architecture?

### 2. The translation

High SKU complexity (size, colour, and club-badge variants) alongside sharp seasonal spikes creates three risks: model drift in reorder tools that miscalculate bulk size-runs around league registration; "shadow AI" — staff pasting client lists or margin figures into public GenAI tools, a direct GDPR breach and pricing leak; and vendor lock-in, where a lean IT team cannot easily walk away from a non-EU-resident cloud. The standing Data Boundary Rule: all raw sales history, ERP tables, and interview recordings are anonymized on-premise before any external API sees them.

### 3. Illustrative use case (CRISP-DM)

**Business understanding:** a reorder forecast must fail safely and visibly, never silently. **Data understanding:** audit where staff already paste client or margin data into public GenAI tools. **Data preparation:** deploy local CLI scripts that strip PII and margin metrics before any API call. **Modelling:** apply tiered governance, from usage-policy guardrails up to human-in-the-loop scrubbing. **Evaluation:** enforce an 85% confidence floor; anything below routes to manual review. **Deployment:** at full maturity, low-confidence forecasts fall back automatically to historical ERP averages.

### 4. Recommended tooling

| Maturity Tier | Safety Mechanism | Wholesaler Use Case |
|---|---|---|
| Tier 1: Off-the-shelf | Usage policies & prompt guardrails | Block public LLMs from answering client queries without verified stock data |
| Tier 2: Workflow automation | Local scrubbing + human-in-the-loop | CLI strips PII/margins; human approves bulk orders |
| Tier 3: Predictive | Automated fallbacks & failure hooks | Confidence <85% defaults to historical ERP averages |

### 5. Responsible AI & trust

Four rules make this operational: local anonymization before any external API; no shadow AI — staff never paste client or margin data into public tools; forecasts below the 85% confidence floor go to manual review, never auto-execute; and failures default to the known-safe historical average, never a guess.

### 6. First five steps for an owner

- Day 1: map staff use of public LLMs across the business.
- Day 1: issue a one-page policy blocking unanonymized ERP uploads to any external AI tool.
- Week 1: deploy local scripts to strip PII and margin metrics before any API processing.
- Week 2: enforce the 85% confidence floor, routing lower-confidence results to manual review.
- Week 3: require EU data residency from AI vendors and verify automated ERP fallbacks.

### 7. sources

Team 6 desk research and operational risk framing carried over from Chapters 1–2.
Regulation (EU) 2024/1689 (EU AI Act); Regulation (EU) 2016/679 (GDPR), eur-lex.europa.eu.
ECI Software Solutions SMB AI survey, ecommercenews.co.nz, 2026.

# Chapter 4: Bridging SME Knowledge Gaps & Technical Readiness

*Why do technically sound AI pilots still fail at SMEs like this one?*

Team 6 · HAN AIBS/AEL 2026 · v1.1 · SME: 100–200-employee Benelux teamwear wholesaler persona · Sources last appraised 24 Oct 2026

### 1. The question, for this kind of SME

**Main Research Question: Why do technically sound AI pilots still fail at regional teamwear wholesalers, and which internal knowledge gaps have to close before a pilot has any chance of succeeding?**

- What makes it hard for a lean, 1–2 person IT team to audit external AI vendor claims?
- How does messy legacy ERP data undermine AI demand forecasting before a model is even chosen?
- What data-literacy and prompting skills do warehouse staff and customer-service reps still lack?

### 2. The translation

Regional wholesalers run 1–2 person IT teams focused purely on ERP uptime, with none of an enterprise's data-engineering bench. Three bottlenecks recur: no structured framework to audit vendor claims, risking overpaying for locked-in SaaS; legacy ERP data with incomplete size-run breakdowns and free-text notes, which drives model drift the moment AI touches it; and staff who know the seasonal spikes but lack the prompting and data-hygiene habits to use agentic tools safely. AI adoption fails less from model limits than from teams lacking the skills to verify AI output against their own ERP records.

### 3. Illustrative use case (CRISP-DM)

**Business understanding:** define what "verified" means before any pilot starts. **Data understanding:** find the order tables with missing size-run or club-badge metadata. **Data preparation:** run local Python CLI scripts that validate and standardize free-text order notes. **Modelling:** train staff on tier-matched prompt templates, from basic hygiene guardrails up to in-house RAG for IT leads. **Evaluation:** run a 14-day parallel test against historical ERP averages before trusting the pilot. **Deployment:** only after that test passes does the business grant automated reorder permissions.

### 4. Recommended tooling

| Maturity Tier | Readiness Focus | Wholesaler Use Case |
|---|---|---|
| Tier 1: Off-the-shelf | Basic data hygiene & prompt safety | Structured templates for club inquiries; zero PII/ERP uploads |
| Tier 2: Workflow automation | Automated data cleaning | Local CLI scripts validate legacy ERP files before forecasting |
| Tier 3: Predictive | Vendor auditing + in-house RAG | IT leads independently verify vendor SLAs and accuracy |

### 5. Responsible AI & trust

Verification over blind trust — every pilot gets a check before it gets trust; zero uploads by default for customer-service staff; vendors are approved only after a four-point check (EU residency, transparent error rates, exportable data, local fallback); and automated reorder permission is granted only after the 14-day parallel test, never on day one.

### 6. First five steps for an owner

- Day 1: identify the top 5 order tables with missing size-run or club-badge metadata.
- Day 1: set standard formatting rules for free-text order notes in the ERP system.
- Week 1: issue a one-page prompting playbook with mandatory verification steps for customer-service reps.
- Week 2: audit external AI vendors against the 4-point checklist.
- Week 3: run the 14-day parallel test before granting automated reorder permissions.

### 7. sources

Team 6 desk research on SME technical readiness gaps; companion to Chapter 5.
ECI Software Solutions SMB AI survey, ecommercenews.co.nz, 2026.
Kolonell, "AI demand forecasting for wholesale," 2026.

## Chapter 5: SME Knowledge Gaps: What Teamwear Wholesalers Still Do Not Know About AI

*What do we still not know about AI, and how do we find out safely?*

Team 6 · HAN AIBS/AEL 2026 · v1.1 · SME: 100–200-employee Benelux teamwear wholesaler persona · Sources last appraised 24 Oct 2026

### 1. The narrow question

For a wholesaler already exposed to the risks mapped in Chapter 3, the question is simple to ask and hard to answer: what do we not know yet about AI, and what would we need to know before committing budget to a tool? Most owners in this sector do not lack opinions about AI — they lack a clear picture of where their own knowledge stops.

### 2. The translation

**Known unknown:** a gap the firm already recognizes, such as not knowing how to validate a vendor's claims before signing — closed by asking the right question. **Unknown unknown:** a gap nobody has named yet, such as discovering only after paying for a forecasting tool that two years of sales history is too inconsistent to use — found only by looking at the firm's own data first.

This chapter maps what is not there yet; Chapter 6 maps what already works. Close the gaps mapped here first, and the use cases on the next page become realistic choices rather than guesses.

### 3. Illustrative use case (CRISP-DM)

**Business understanding:** reduce stockouts and overstock on seasonal club-kit size-runs. **Data understanding:** pulling ERP sales history to feed a forecasting model usually surfaces missing SKU fields or inconsistent product codes nobody flagged beforehand — a known unknown turning into an unknown one. **Data preparation:** cleaning that data costs real, unbudgeted labor; get the vendor's minimum clean-history requirement in writing before signing. **Modelling:** a generic model applied without that cleanup is exactly how the Chapter 3 drift risk shows up in practice. **Evaluation:** the Tier 3 confidence floor from Chapter 3 becomes practical here, flagging low-confidence forecasts for manual review. **Deployment:** deploying before the gap closes means deploying on guesswork — the knowledge gap, not the tool, is usually the real blocker.

### 4. Recommended tooling

- **Staff AI-use survey:** informal, covers ad hoc tools like ChatGPT management may not know staff already use.
- **Readiness self-assessment:** free AI readiness checklists, often available via regional SME support programs.
- **ERP data-quality spot check:** a small random SKU sample, checked for missing or inconsistent fields, before any vendor conversation.

### 5. Responsible AI and Trust

The Chapter 3 Data Boundary Rule applies here too, even though this stage feels informal: no raw ERP exports, sales records, or client-naming survey comments go into any external tool while mapping these gaps.

### 6. First five steps for an owner

- Map current tool use: list, informally, which AI tools staff already use.
- Name one trust condition: pick one pain point and write down what would need to be true to trust AI with it.
- Spot-check the ERP: pull 20 SKU records and check for missing or inconsistent fields.
- Ask an outside source: an accountant or SME support desk, what comparable firms actually gained from AI.
- Write the gap list down formally: direct input for choosing a use case in Chapter 6.

### 7. sources

Internal: Chapter 3, Trust and AI Safety for Teamwear Wholesalers.
ECI Software Solutions SMB AI survey, ecommercenews.co.nz, 2026.
Australian Government, AI adoption insights, ai.gov.au, Dec 2025–Feb 2026; Kolonell, 2026.

# Chapter 6: Practical AI Use Cases for Regional Teamwear Wholesalers

*Which AI use cases are actually proven and worth the investment?*

Team 6 · HAN AIBS/AEL 2026 · v1.1 · SME: 100–200-employee Benelux teamwear wholesaler persona · Sources last appraised 24 Oct 2026

### 1. The question, for this kind of SME

**Main Research Question: Which AI use cases are actually proven and worth the investment for a regional teamwear wholesaler, as opposed to vendor hype?**

- Which AI use cases have published, audited evidence behind them, rather than marketing claims alone?
- How do global AI strategies in the US, China, and Europe differ in ways that matter for an SME?
- What operational KPI should anchor a decision to invest at each maturity tier?

### 2. The translation

SMEs in regional manufacturing and teamwear wholesale (100–200 staff, 1–2 IT) face broken size runs, manual PIM updates, high fabric scrap, and long cut-to-ship cycles. Rather than enterprise-grade AI, they need practical, modular use cases across three maturity tiers, ranked by published rather than projected impact.

### 3. Illustrative use case (CRISP-DM)

**Business understanding:** pick the next tier to invest in based on audited outcomes, not projections. **Data understanding:** gather comparative benchmarks across all three tiers — case studies, vendor data, real operational KPIs. **Data preparation:** separate marketing claims from audited figures. **Modelling:** rank by realistic impact — Tier 1 cuts physical sample requests; Tier 2's automated nesting case studies report roughly 1% fabric-scrap reduction, well below some vendor claims; Tier 3 raised product availability from 90% to 97% in one real apparel-retailer case (Kiabi, via Infor Retail Demand Management). **Evaluation:** weigh each tier's return against the wholesaler's own KPI focus. **Deployment:** commit capital in tier order, each gated by evidence, never leapfrogging to Tier 3 on hype.

## 4. Recommended tooling

| Maturity Tier | Use Case | Operational Impact |
|---|---|---|
| Tier 1: Off-the-shelf | AI visual/3D product preview (e.g. Tricorp TriFit) | Cuts physical sample requests, faster B2B quotes |
| Tier 2: Workflow automation | Automated PIM/ERP enrichment, pattern nesting (Lectra) | ~1% fabric-scrap reduction, less manual re-entry |
| Tier 3: Predictive | Demand forecasting & stock replenishment | 90% → 97% product availability (Kiabi case) |

## 5. Responsible AI & trust

Global AI strategies diverge: the US is capital-intensive and platform-driven; China runs state-directed, vertically integrated manufacturing — Shein compresses design-to-production to 3–7 days versus 3–4 weeks in traditional fast fashion (Sacra); Europe prioritizes human-in-the-loop governance and GDPR/EU AI Act compliance. This handbook follows Europe's route: explicit monitoring hooks catch drift before it reaches a client workflow, the 1–2 person IT team is trained to manage prompts and exceptions itself, and data stays on regional servers under scheduled EU AI Act review.

## 6. First five steps for an owner

- Weeks 1–2: audit ERP logs for broken size-run patterns; set your primary KPI focus.
- Weeks 1–2: run local data-anonymization checks on customer and employee identifiers.
- Weeks 3–4: pilot a 3D preview tool for B2B quotes and automated PIM for SKU variants.
- Weeks 5–6: set monitoring hooks for drift/confidence drops; train IT and warehouse leads.
- Weeks 5–6: review EU AI Act compliance and confirm regional data residency.

## 7. sources

Tricorp Workwear, TriFit product page, tricorp.com.
Sacra, research note on Shein's manufacturing model, sacra.com; Lectra automated-nesting case studies, lectra.com.
Regulation (EU) 2024/1689 (EU AI Act); Kiabi case study via bestpractice.ai, citing solutions-numeriques.com.

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*Disclaimer: this handbook (Chapters 1–6) is applied classroom research produced by a HAN University of Applied Sciences student team and is not professional, legal, or financial advice. Verify every recommendation against your own data, vendor contracts, and applicable regulations before acting on it.*

*Note on length: the course requires one page per chapter across the six weekly chapters, which produces six pages in total; it was not possible to also meet the separately stated 4-page overall maximum without cutting weekly chapters entirely.*